Tracking 3-D Motion of Dynamic Objects Using Monocular Visual-Inertial Sensing

Kejie Qiu, Tong Qin, Wenliang Gao, and Shaojie Shen

***Abstract*—Six degree-of-freedom (6-DoF) visual tracking of dynamic objects is fundamental to a large variety of robotics and augmented reality (AR) applications. A key to this problem is accurate distance measurement of dynamic objects, which is usually obtained via stereo cameras, RGB-D sensors, or LiDARs. In this paper, however, we address the problem using only a monocular camera rigidly mounted with a low-cost inertial measurement unit. This is a light-weight, small-size, and low-cost solution, which is particularly suitable for tracking dynamic objects on drones or on mobile phones. Starting from a generic image-based two-dimensional tracker, we propose a novel method to resolve the object scale ambiguity in monocular vision in a geometric manner based on correlation analysis. This enables accurate metric three-dimensional tracking of arbitrary objects without requiring any prior knowledge about the object shape or size. We discuss the applicability by analyzing the observability condition and degenerated cases for object scale recovery. Simulation and real-world experimental results with ground truth comparison, along with AR application examples, demonstrate the feasibility of the proposed 6-DoF tracking method.**

***Index Terms*—Computational geometry, three-dimensional (3-D) perception, simultaneous localization and mapping (SLAM), visual tracking.**

I. Introduction

A COMPLETE three-dimensional (3-D) perception system, including ego-motion estimation (localization), static environment mapping, and dynamic object tracking, has large application potential for microaerial vehicles (MAVs), mobile robots, self-driving, and augmented reality (AR) development. Obviously, two-dimensional (2-D) object tracking on the image plane is not enough for these real-world applications involving 3-D object perception. In other words, the full six degree-of-freedom (6-DoF) pose estimation of the tracked dynamic object is needed. In order to achieve accurate 6-DoF dynamic object tracking, stereo camera [1] and Lidar [2] are often used since the scale of the tracked object is directly measured, whereas we are particularly interested in achieving 3-D object tracking only using the minimal sensor suite that consists of only a monocular camera and an inertial measurement unit (IMU), due to its superior size, weight, and power characteristics. It is also widely available in mobile devices and suitable for large-scale deployment. With 3-D perception of dynamic objects, a drone can actively track a mobile vehicle by predicting the vehicle trajectory. In addition, scene AR can be extended to dynamic AR, which can render virtual objects onto the dynamic objects instead of the static scene. Two potential application scenarios, active tracking using a drone and dynamic AR are shown in Fig. 1.

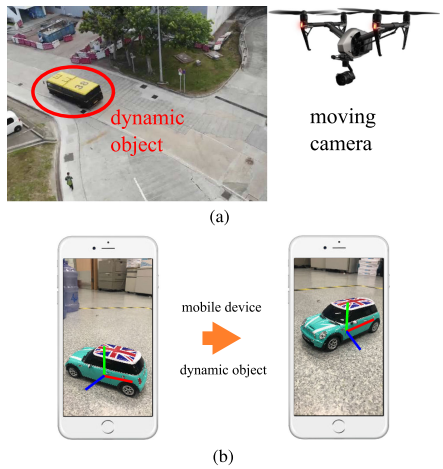


Fig. 1. Potential applications with 6-DoF tracking of dynamic objects using monocular visual-inertial sensing (photos from Internet). (a) Active tracking with a drone. (b) Dynamic AR with a handheld device.

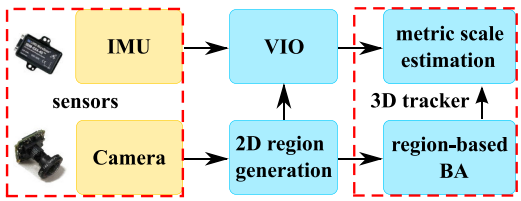


Fig. 2. Overall system structure. The minimal sensor suite consists of a monocular camera and an IMU is used for perception.

As Wang *et al.* [2] pointed out, simultaneous localization and mapping (SLAM) and dynamic object tracking are mutually beneficial. In fact, our 3-D object tracking system is built on accurate ego-motion estimation. As shown in Fig. 2, our system starts with a moderate 2-D object region generation module, from which a region-based bundle adjustment (BA) is performed

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on the tracked dynamic object utilizing the 2-D object regions. Up-to-scale camera poses in the object frame are recovered from this step. Since the object is regarded as the reference frame in the region-based BA, we assume that the tracked object is rigid. On the other hand, the image backgrounds masked by the object regions and the IMU data are used for scaled camera pose estimation in the world frame using a monocular visual-inertial odometry (VIO) method. Finally, we present a novel motion correlation analysis-based scale recovery method to estimate the metric scale of the object and recover the full 6-DoF object pose in the world frame. In addition, the ego-motion estimation can be more accurate since the moving parts will be filtered out according to the dynamic object region generation results.

The metric scale of camera poses in the world frame (inertial reference frame in physics) is directly observable using monocular visual-inertial sensing because the onboard IMU can be used as the metric scale source in the VIO. However, the biggest challenge of 3-D object tracking using monocular visual-inertial sensing is that the object scale cannot be measured directly. This is because the dynamic object is regarded as the reference frame in the region-based BA, and the object frame is not even an inertial reference frame in physics, which means the IMU that is rigidly mounted with the camera cannot be used to constrain the object scale directly. Considering that the metric scale of camera poses in the VIO is well-observable, we focus on the metric pose recovery of the dynamic object, especially the metric scale estimation of the dynamic object. Unless stated, "scale" refers to the metric scale of the tracked dynamic object in the following.

Most state-of-the-art object tracking methods using monocular vision are model-based [3] or learning-based [4], [5]. These methods first recognize the tracked object and then recover the object scale according to a predefined or pretrained object prior. However, this may cause scale ambiguity in some scenarios. For example, the similarity in appearance between a real car and a toy model car may lead to incorrect scale estimation if only the previous information of a car is used. On the other hand, trajectory reconstruction adopts an object motion assumption to recover the object trajectory from monocular observations [6], [7]. They are actually trajectory fitting methods that utilize multiple object motion assumptions to solve the metric scale of the object trajectory. For example, the motion assumption can be planar conic motion [8] or conserved motion [9]. Hence, the performance of these methods highly relies on the motion model selection for a certain object.

Our method is different from the above methods since neither the object scale prior nor specific motion assumption of the tracked object is needed. We are inspired by the idea from trajectory reconstruction of introducing an additional constraint (motion assumption) to make the metric scale solvable, but we make use of the universal fact that the camera motion and the object motion are independent as the additional constraint to solve the metric scale estimation problem. We find that the motion independence is a more general and reasonable property compared with the motion assumptions used in trajectory reconstruction. Mathematically, independence means a statistically uncorrelated tendency. If we properly quantify the correlation between the camera motion and the object motion and formulate an objective function based on the quantification result, we can recover the metric scale of the object by minimizing the objective function. The object pose in the world frame can also be recovered.

To be specific, we solve the metric scale estimation problem from the perspective of signal processing: the camera motion and object motion in the world frame can be regarded as two isolated discrete-time random processes. This process is similar to the correlation-based signal decomposition discussed in the telecommunication community [10]. The idea of metric scale estimation using correlation analysis was first proposed in our previous work [11]. This paper significantly improves on our previous work in the following ways.

1) We implement a more robust 2-D object region generation module, in which 2-D object detection is used for 2-D object tracking initialization. In our previous work, the 2-D tracking initialization relies on manual region selection of the object.
2) By using a more reliable correlation quantification method (trace correlation), we now achieve more accurate and robust scale estimation compared to the previous method. In addition, the corresponding observability condition can be softened to some extent, resulting in an increased scale acceptance rate.
3) In this paper, we build a simulation environment for comprehensive degenerated case validation and show a comparison of different objective function formulation methods including the one from our previous work. In that work, several common degenerated cases are discussed through real tracking experiments.

We identify the contributions of this paper as threefold.

1) We propose a geometry-based 3-D tracking system with metric scale awareness for generally rigid objects using monocular visual-inertial sensing. Neither the object scale prior nor motion assumption of the dynamic object is needed.
2) We complete the 3-D tracking system by estimating the metric scale of the tracked object based on trace correlation analysis between the camera motion and object motion.
3) We study the key observability condition of the proposed metric scale estimation method, such that multiple degenerated cases can be detected and rejected.

The rest of this paper is structured as follows. Section II introduces relevant work on 6-DoF object pose estimation and fundamental work that our system relies on. In Section III, our system structure and the major frame relationships are illustrated. Section IV introduces the 2-D region generation module, and Section V introduces how the region-based BA works. The key motion statistics used for objective function formulation are introduced in Section VI. In Section VII, we present a robust objective function formulation method based on trace correlation and discuss the corresponding observability condition. The limitations of our tracking system are discussed in Section VIII. Section IX shows the simulation results of different objective formulation methods for typical motion styles of the camera and the object; the degenerated cases of the metric scale module are also analyzed in the simulation environment. Section X gives the experimental results with comparison against the ground truth. Section XI concludes this paper.

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II. Related Work

Six-DoF object tracking is a well-studied problem with dedicated sensor suites. One common solution is to utilize multiple fixed and synchronized cameras such that the triangulation constraint is still valid even when the objects are moving. Motion capture systems such as OptiTrack[1] or Vicon[2] are a mature object tracking technology for moderately sized application scenarios, but reflective markers have to be attached to the tracked objects. Besides the commercial active-tracking systems with specialized markers, passive-tracking systems for sports scene analysis and traffic surveillance have also been proposed since the targets are usually noncooperative in these cases. A trajectory estimation system for MAVs using flight dynamics as a prior is proposed in [12] based on fixed ground cameras, assuming that all the cameras are synchronized, whereas Vo *et al.* [13] proposed a spatiotemporal BA framework to simultaneously estimate the temporal alignment between cameras. Zou and Tan [14] proposed CoSLAM, a SLAM system in dynamic environments with multiple cameras mounted on moving platforms, where the trajectories of moving foreground points are also recovered. However, all of these methods are limited by the fixed-camera or multiple-camera setting.

Single sensor-based methods using an RGB-D sensor [15] or a stereo camera [16], [17] can relax the multisensor condition at the cost of a limited sensing range. Consequently, these methods can only handle small-scale application situations. To overcome the sensing limitation, researchers prefer to use Lidars for reliable object detection and tracking. Wang *et al.* [2] presented a SLAM algorithm with generalized objects with a joint optimization style, and developed another algorithm in which the stationary part and moving objects are estimated separately. In addition, many autonomous driving systems utilize Lidars to achieve detection and tracking of typical moving objects, such as pedestrians, bicycles, vehicles, etc. [18], [19]. These methods are all based on depth sensors such that the object scale can be inferred from the depth observation.

Few methods adopt a monocular camera to recover the 6-DoF pose of dynamic objects, since the tracking results may lack metric scale unless the tracked object is studied or modeled in advance. Model-based tracking is one branch of monocular camera-based methods. Given a set of known 2-D-to-3-D correspondences, the relative pose can be solved with Perspective-n-Point [3] or view alignment using edge features [20], [21]. For instance, ARUCO [22] makes use of predesigned tags as the known 3-D information to mark tracking targets and calculates the camera poses according to the tag observations in the captured images. Instead of using predesigned landmarks, discriminative feature points such as BRISK [23] are also practical for 2-D-to-3-D matching. In addition to sparse point features, dense method [24] and edge-based method [25] are good alternatives for textureless object tracking. Another significant branch is learning-based tracking [4]. Pavlakos *et al.* [5] proposed an efficient convolutional network to first locate reliable predefined semantic keypoints and then estimate the 6-DoF pose with only a monocular camera. However, the similarity in appearance of objects with different sizes may lead to incorrect scale estimation if only a pose detection network is employed because of the lack of metric scale measurement. The use of preconstructed 3-D model and trained neural network means that the scale of a certain object is encoded in advance. In contrast, our geometry-based 3-D tracking method has the metric scale awareness ability for generally rigid objects without an object scale prior.

Even though monocular visual-inertial fusion-based state estimation [26], [27] and dense environment mapping [28], [29] have shown promising results, metric 6-DoF pose tracking of dynamic objects is still a challenge with such limited sensing. In this paper, we propose to estimate the metric scale by analyzing the statistical correlation between the camera motion and the recovered object motion. We utilize VINS [26] for camera motion estimation such that the object motion can be projected from the camera frame to the world frame. The up-to-scale object motion in the camera frame is estimated by using an object region-based BA. To obtain accurate object regions for region-based BA, a robust 2-D tracker is needed. The 2-D tracking problem has been well-discussed and many classic tracking methods such as CMT [30] and STRUCK [31] are robust for common cases, resulting in sequential 2-D bounding boxes of the tracked object. The latest 2-D tracking methods based on deep learning can provide even better performance [32]. Given accurate object regions on sequential images, a region-based BA is performed to obtain up-to-scale 6-DoF object pose. Multibody structure-from-motion [33] extends the classical BA framework to dynamic scenes with multiple moving objects, in which multiple region-based BA threads are carried out for the moving objects. But this method only corrects the relative object scales rather than absolute metric scales.

Finally, our metric scale estimator is inspired by a problem called trajectory reconstruction, in which by introducing an additional constraint in the time domain, the object scale is solved with monocular observations. This problem is studied to handle trajectory reconstruction of a point target from a monocular image sequence. Avidan and Shashua [8] solved straight-line and planar conic motions, whereas Yu *et al.* [6] and [7] extended it to general motion cases with polynomial fitting, in which the trajectory is represented by a linear combination of polynomial bases. Park *et al.* [34] made use of discrete cosine transform basis as the object trajectory bases. These fitting-based methods utilize the motion assumption constraints but highly rely on fitting model selection for good scale estimation. They try to estimate multiple object depths in all image frames instead of the intrinsic scale of the tracked object. In addition, the tracked object is regarded as a point, and the rotation of the object is not estimated. Our method, on the other hand, recovers the whole 6-DoF object pose without any object motion priors.

III. System Overview

As shown in Fig. 2, several independent modules comprise the proposed system, which uses the minimal sensing suite for

[1][Online]. Available: http://optitrack.com/

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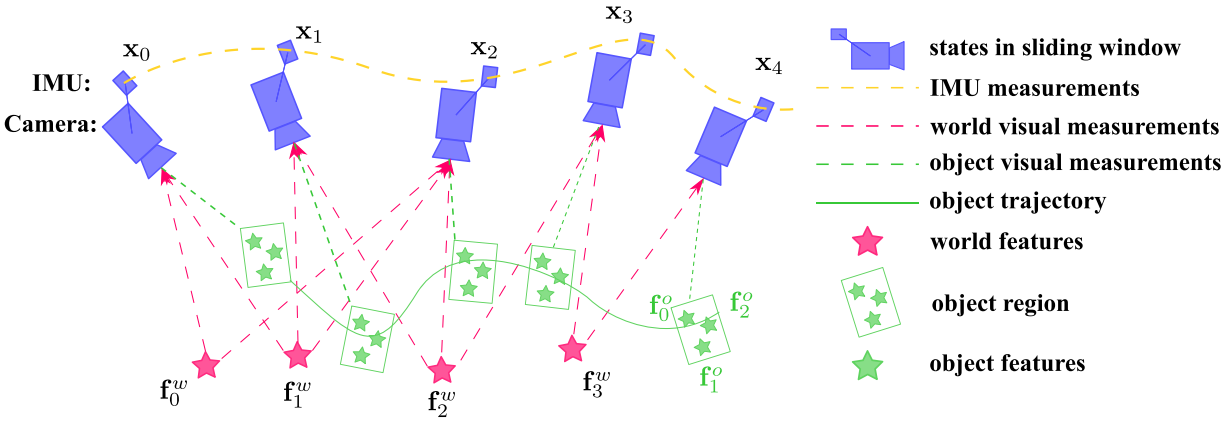


Fig. 3. Unified illustration of the sliding window structures for both VIO and region-based BA in the world frame. Five camera states together with four fixed environment features and three dynamic object features are depicted.

perception. The main purpose is to recover 3-D object motion from 2-D object tracking results using monocular visual-inertial sensing. To be specific, the 2-D object region generation module is achieved by 2-D object tracking, which uses 2-D object detection for tracking initialization. The VIO module is implemented by VINS [26]. In this paper, we focus on the region-based BA module and the metric scale estimation module. Both VINS and region-based BA are formulated as sliding window-based nonlinear optimization problems, and a unified representation of the sliding window structure is shown in Fig. 3. Since the object is regarded as the reference frame in the region-based BA, the tracked object must be a rigid one, which is the basic assumption of our tracking system. Besides that, we do not need to know as to which type of tracked object it is. Nor do we need to have motion prior assumptions about the object. For example, we can track and recover the 3-D motion for both a real car and a toy model car, although they have a similar appearance but different sizes. In addition, what we estimate is a time-invariant metric scale that reflects the intrinsic scale of the tracked object. It means the metric scale estimation module need not work all the time.

A more detailed working pipeline and the frame relationship of our 3-D tracking system is shown in Fig. 4. First of all, the object regions on the captured images are generated by the 2-D object region generation module. Given consecutive image regions of the tracked object, the VIO takes both the IMU and masked image data for scaled camera pose estimation in the world frame. Meanwhile, the region-based BA module that takes the object regions as input is simultaneously running for up-to-scale camera pose estimation in the object frame. Both motion data from the VIO and region-based BA are sent to the metric scale estimation module, where the object motion will be projected to the world frame for motion correlation analysis. Once the collected motion data are sufficient for estimation, the scale of the tracked object will be recovered by motion correlation analysis with an observability condition check.

For frame definition, $(\cdot)^w$ is the world frame, $(\cdot)^c$ the camera frame, and $(\cdot)^o$ the object frame. The IMU frame is ignored since it only directly constrains the metric scale of the VIO. $\mathbf{p}_y^x$ and $\mathbf{R}_y^x$ are the 3-D translation and rotation of frame $(\cdot)^y$, respectively, with respect to frame $(\cdot)^x$. The camera pose in the world provided by the VIO is represented by $\mathbf{R}_c^w$ and $\mathbf{p}_c^w$, and $\bar{(\cdot)}$ denotes the up-to-scale pose results. $\mathbf{R}_o^c$ and $\bar{\mathbf{p}}_o^c$ together denote the relative transformation of the object frame with respect to the camera frame provided by the region-based visual BA module. What the metric scale estimation module estimates is the scale ratio s that converts the up-to-scale position into a scaled one ($\mathbf{p}_o^c = s \cdot \bar{\mathbf{p}}_o^c$). We use $\mathbf{R}_o^w$ and $\mathbf{p}_o^w$ together to denote the global transformation from the world frame to the object frame. All the estimated values are denoted by $\hat{(\cdot)}$, and the estimated object position in the camera frame can be represented as follows:

$$\hat{\mathbf{p}}_{o_t}^{c_t} = \hat{s} \cdot \hat{\bar{\mathbf{p}}}_{o_t}^{c_t}. \tag{1}$$

Finally, the 6-DoF object pose in the world frame is recovered using the estimated metric scale through rigid body transformation

$$\begin{aligned} \hat{\mathbf{R}}_{o_t}^w &= \hat{\mathbf{R}}_{c_t}^w \hat{\mathbf{R}}_{o_t}^{c_t} \\ \hat{\mathbf{p}}_{o_t}^w &= \hat{s} \cdot \hat{\mathbf{R}}_{c_t}^w \hat{\bar{\mathbf{p}}}_{o_t}^{c_t} + \hat{\mathbf{p}}_{c_t}^w. \end{aligned} \tag{2}$$

IV. 2-D Object Region Generation

The 2-D object region generation module works as the key frontend for both region-based BA and VIO, as shown in Fig. 4. The object regions of the captured images are for the region-based BA module, where the object will be regarded as the reference frame for up-to-scale 6-DoF camera pose estimation in the object frame, whereas the image backgrounds masked by the object regions will be used by the VIO. The object region generation module in our system consists of a 2-D object detector and a 2-D object tracker, where the detector works as the object region initialization for the 2-D tracker. If multiple-object tracking is needed, several independent 2-D tracking threads will be running simultaneously accordingly, and the corresponding object regions will be processed by independent region-based BA threads, where the camera poses in different object reference frames will be estimated simultaneously.

There are diverse approaches to represent object regions. One common way is to use 2-D bounding boxes that are the direct

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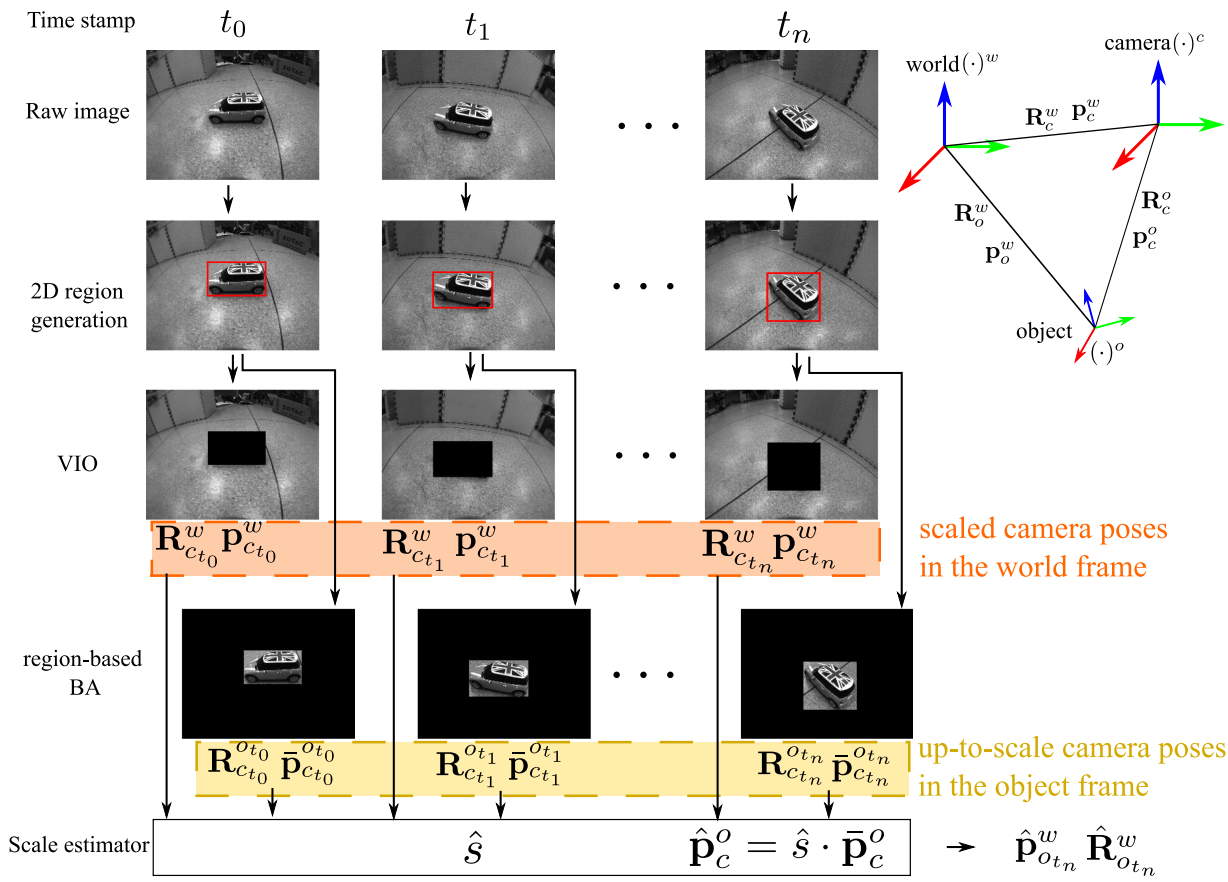


Fig. 4. System pipeline and coordinate frames. The sequential images from timestamps t_0 to t_n are shown for tracking a toy model car. The generated 2-D object regions are used for decomposition of the background regions and object regions, which are the input data for the VIO and the region-based BA, respectively. The VIO also uses the IMU data as input for scaled pose estimation, while the region-based BA only outputs up-to-scale pose estimation. Both pose estimations are sent to the metric scale estimator, once the collected pose data is sufficient for scale estimation (at timestamp t_n). Then the intrinsic object scale will be estimated and the metric 6-DoF object pose can be recovered.

results of traditional object detection and tracking methods. Although several background pixels are included by using bounding boxes, the region-based BA is robust against such outliers as long as the object region is dominant in the bounding box. A learning-based object detector and tracker is another choice. For example, the recent work mask R-CNN [35] can generate pixel-wised segmentation results, which can achieve the maximum level of outlier elimination in the object region generation phase. In order to achieve real-time performance, we adopt YOLO [36] as the 2-D detector and Re-3 [32] as the 2-D tracker in our experiments, where 2-D bounding boxes are used for object region representation.

V. Up-to-Scale Region-Based BA

The target of this module is to recover 3-D object motion from 2-D object regions of sequential monocular images captured by a moving camera. This is quite different from the 2-D tracking that aims to recover the object regions in images. In order to extend 2-D object tracking to 3-D object tracking, an object region-based BA is designed to generate up-to-scale object pose estimation with respect to the camera. Since region-based BA takes the object regions of the captured images as input, the object can be either static or dynamic for the BA initialization and following adjustment process. The frontend of the region-based BA adopts the KLT sparse optical flow [37] for corner feature association, followed by an outlier rejection step using RANSAC with a fundamental matrix model [38].

Note that this BA is performed with respect to the object frame; that is, the 3-D tracker estimates the camera motion w.r.t. the object frame, which is different from the visual BA defined in the VIO, which estimates the camera motion w.r.t. the world frame. Since both the camera frame and the object frame are dynamic, either camera motion or object motion can cause the relative motion estimated by the region-based visual BA. That is to say, the estimated motion of the region-based BA is a compound motion combining camera and object motion.

The full state of a sliding window with N camera frames and M object features is defined as follows:

$$\begin{aligned} \mathcal{X} &= [\mathbf{x}_0, \mathbf{x}_1, \ldots \mathbf{x}_{N-1}, \mu_0, \mu_1, \ldots \mu_{M-1}] \\ \mathbf{x}_k &= \left[\bar{\mathbf{p}}_{o_k}^{c_k}, \mathbf{q}_{o_k}^{c_k}\right], k \in [0, N-1] \end{aligned} \tag{3}$$

where the kth camera state consists of the up-to-scale position $\bar{\mathbf{p}}_{o_k}^{c_k}$, and orientation $\mathbf{q}_{o_k}^{c_k}$ of the camera frame c_k w.r.t. the object frame o_k. 3-D features are parameterized by their inverse

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depth μ when first observed in the camera frame. The BA can be formulated as a nonlinear least-squares problem

$$\min_{\mathcal{X}} \sum_{(l,j)\in\mathcal{C}} \left\|\mathbf{r}_{\mathcal{C}}(\hat{\mathbf{z}}_l^{c_j}, \mathcal{X})\right\|_2^2 \tag{4}$$

where $\mathbf{r}_{\mathcal{C}}(\hat{\mathbf{z}}_l^{c_j}, \mathcal{X})$ is the nonlinear residual function of the visual measurements. The vector $\hat{\mathbf{z}}_l^{c_j}$ results from the procedure in which the lth feature is projected to the camera frame c_j. The visual residual is defined as the sum of the reprojection errors between the projected 3-D features and the observed 2-D features. Consider the lth feature that is first observed in the ith image, the residual for the feature observation in the jth image is defined as

$$\begin{aligned} \mathbf{r}_{\mathcal{C}}\left(\hat{\mathbf{z}}_l^{c_j}, \mathcal{X}\right) &= \left(\begin{bmatrix} \hat{u}_l^{c_j} \\ \hat{v}_l^{c_j} \end{bmatrix} - \pi_c(\mathcal{P}_l^{c_j})\right) \\ \mathcal{P}_l^{c_j} &= \left(\mathbf{R}_o^{c_j}\left(\mathbf{R}_{c_i}^{o}\frac{1}{\lambda_l}{\pi_c}^{-1}\left(\begin{bmatrix} \hat{u}_l^{c_i} \\ \hat{v}_l^{c_i} \end{bmatrix}\right) + \mathbf{p}_{c_i}^{o}\right) - \mathbf{p}_{c_j}^{o}\right) \end{aligned} \tag{5}$$

where $[\hat{u}_l^{c_i}, \hat{v}_l^{c_i}]$ is the first observation of the lth feature in the ith image, $[\hat{u}_l^{c_j}, \hat{v}_l^{c_j}]$ is the observation of the same feature in the jth image, π_c is the camera projection function, and π_c^{-1} is the back projection function that turns a pixel location into a unit plane using camera's intrinsic parameters.

The direct optimization results are camera transformations w.r.t. the object frame, but what we need for 3-D object tracking is the object transformation. Thus, the object coordinates have to be located on the object itself, which is not guaranteed by a regular visual BA framework because the coordinates of the reference frame are set to be identical to the initial camera coordinates when the BA finishes initialization. This coordinate initialization method is feasible for visual SLAM systems since the world coordinates can be anywhere for correct odometry representation. For 3-D object tracking, however, since the object is dynamic and the original point of the object coordinates represents the object location, the object coordinates must be located on the object described by the reconstructed object points.

To solve this problem, we further modify the object coordinates by estimating the initial object pose in the camera frame when the region-based BA finishes initialization. Since the absolute object orientation is unknown, we set the object orientation as an identity rotation matrix when the region-based BA is initialized, and estimate the relative 3-D rotation during tracking. As shown in Fig. 5, we move the object coordinates from the initial position to the object points along the direction denoted by the 2-D object region center, then normalize the object center depth to 1 by scaling all the object points simultaneously. In fact, the object coordinates can be anywhere around the object points. Once it is determined in the coordinate modification process, it will be continuously tracked in the BA framework.

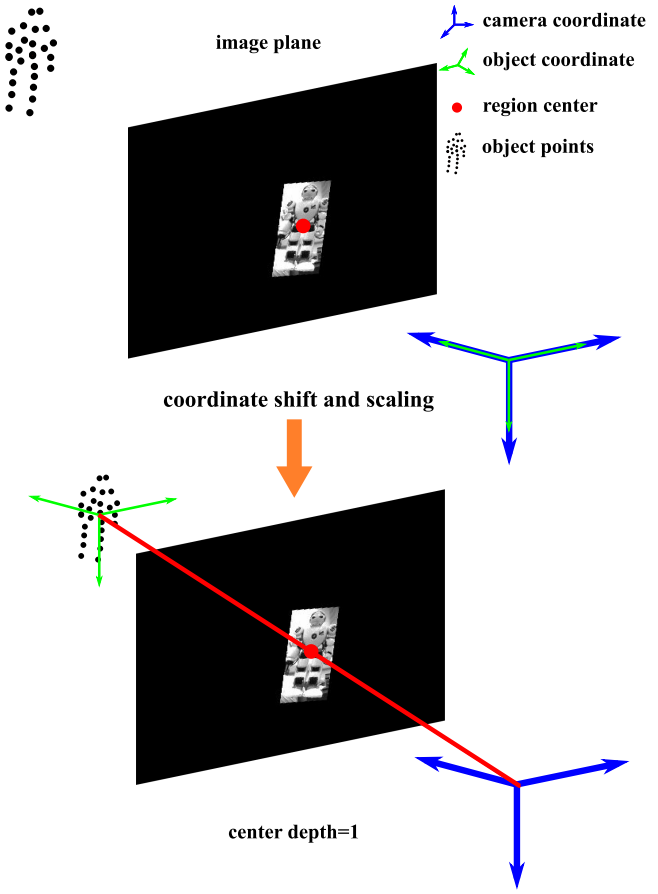


Fig. 5. Object coordinate modification process. The object coordinate will be attached with the object points and the depth of the region center will be normalized to 1 after modification.

VI. Motion Statistics

Before performing metric scale estimation, we already have accurate camera poses in the world frame provided by the VIO, and up-to-scale object poses in the camera frame provided by region-based BA (see Section V). The metric scale of the tracked object is conditionally solvable if the camera motion and the object motion satisfy the scale observability condition, which will be discussed later (see Section VII-C). The object scale estimation problem in this paper is solved from the perspective of statistics and probability. As we know that the direct observation from the region-based BA is a compound motion of the object motion and camera motion, while the object motion and camera motion in the world frame are prone to be independent, there is a tendency that the object motion and camera motion in the world frame will be statistically uncorrelated. Thus, the metric scale can be inferred from this uncorrelation property.

We start by introducing the scale factor (s) we want to estimate. It is actually the scale ratio of the real metric scale over the inherent scale that exists in the region-based BA

$$\mathbf{p}_{o_t}^{c_t} = s \cdot \bar{\mathbf{p}}_{o_t}^{c_t}. \tag{6}$$

Since the initial camera pose and object points are normalized in the region-based BA, the value of the scale ratio equals the real initial depth between the camera and the object at the moment the region-based BA is initialized. It is time-invariant since it reflects the intrinsic scale of the tracked object. Therefore, we can accumulate a period of motion observations, waiting for the opportunity that the observability condition is satisfied to estimate this scale ratio. Once the scale ratio is estimated, it can

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work with the region-based BA to provide continuous metric pose tracking. We refer to this scale ratio as "scale" in the following for simplicity. Accordingly, the estimated scale $\hat{s}$ can be represented as the sum of the true scale s and the estimation error Δs

$$\hat{s} = s + \Delta s. \tag{7}$$

We define the rotation compensated object position as $\mathbf{p}_{d_t}^w$

$$\mathbf{p}_{d_t}^w = \mathbf{R}_{c_t}^w \bar{\mathbf{p}}_{o_t}^{c_t}. \tag{8}$$

Recall the recovered object trajectory (2)

$$\hat{\mathbf{p}}_{o_t}^w = \hat{s} \cdot \mathbf{R}_{c_t}^w \bar{\mathbf{p}}_{o_t}^{c_t} + \mathbf{p}_{c_t}^w = \hat{s} \cdot \mathbf{p}_{d_t}^w + \mathbf{p}_{c_t}^w \tag{9}$$

particularly, when $\hat{s}$ equals s

$$\mathbf{p}_{o_t}^w = s \cdot \mathbf{p}_{d_t}^w + \mathbf{p}_{c_t}^w \tag{10}$$

which means $\mathbf{p}_{d_t}^w = (\mathbf{p}_{o_t}^w - \mathbf{p}_{c_t}^w)/s$, so the physical meaning of $\mathbf{p}_{d_t}^w$ is the up-to-scale position difference of the object position and the camera position in the world frame.

The recovered object trajectory has two components

$$\begin{aligned} \hat{\mathbf{p}}_{o_t}^w &= \hat{s} \cdot \mathbf{p}_{d_t}^w + \mathbf{p}_{c_t}^w = \mathbf{p}_{o_t}^w + \Delta s \cdot \mathbf{p}_{d_t}^w \\ &= \left(1 + \frac{\Delta s}{s}\right) \mathbf{p}_{o_t}^w - \frac{\Delta s}{s} \mathbf{p}_{c_t}^w \end{aligned} \tag{11}$$

and the second component is the camera trajectory in the world frame. It means that an incorrect metric scale estimate ($\Delta s \neq 0$) will lead to insufficient trajectory decomposition. The final recovered object trajectory will be a mixture of an incorrect object trajectory component and a nonzero camera trajectory component. Therefore, the recovered object trajectory will be highly correlated with the camera trajectory. On the other hand, the correct scale estimate ($\Delta s = 0$ or $\hat{s} = s$) will remove the camera trajectory component from the recovered object trajectory. The correlation between the recovered object trajectory and the camera trajectory becomes the correlation between the true object trajectory and the camera trajectory. The true object trajectory and the camera trajectory are weakly correlated, which will be further guaranteed by the designed observability condition. In fact, the observability condition is used to avoid multiple degenerated cases in which the scale cannot be correctly recovered.

Thus, the object scale can be estimated by a correlation analysis between the recovered object trajectory and the camera trajectory in the world frame. If the object trajectory and camera trajectory satisfy the observability condition within a specific duration, we argue that a good scale estimate ($\Delta s \to 0$ or $\hat{s} \to s$) can be obtained by minimizing the quantified correlation between the recovered object trajectory and the camera trajectory in the world frame.

In practice, we extract motion features from the corresponding trajectories to make the metric scale estimator more robust against the accumulated drift of the VIO and the region-based BA. The motion feature is a certain order derivative of the corresponding trajectory. Another advantage of using the motion feature is that it introduces more randomness than using the trajectory (derivative is sensitive to local changes), which enables us to estimate the metric scale with relatively few motion observations. Basically, the motion feature is time-sequential data. Therefore, we model the recovered object motion and camera motion as two discrete-time random processes $\hat{\mathbf{m}}_o(t)$ and $\mathbf{m}_c(t)$, and define another random process $\mathbf{m}_d(t)$ to denote the up-to-scale motion difference

$$\begin{aligned} \mathbf{m}_c(t) &= [m_c^x(t), m_c^y(t), m_c^z(t)]^T = \frac{\partial^k \mathbf{p}_{c_t}^w}{\partial t^k} \\ \mathbf{m}_d(t) &= [m_d^x(t), m_d^y(t), m_d^z(t)]^T = \frac{\partial^k \mathbf{p}_{d_t}^w}{\partial t^k} \\ \hat{\mathbf{m}}_o(t) &= [\hat{m}_o^x(t), \hat{m}_o^y(t), \hat{m}_o^z(t)]^T = \frac{\partial^k \hat{\mathbf{p}}_{o_t}^w}{\partial t^k} \\ &= \hat{s}\mathbf{m}_d(t) + \mathbf{m}_c(t) = \left(1 + \frac{\Delta s}{s}\right) \mathbf{m}_o(t) - \frac{\Delta s}{s} \mathbf{m}_c(t) \end{aligned} \tag{12}$$

where $\mathbf{p}_{d_t}^w$ is the rotation compensated object position which is defined in (8).

We directly approximate the motion features according to the raw data, taking the first-order derivative as an example

$$\begin{aligned} \mathbf{m}_c(t) &= \frac{\partial \mathbf{p}_{c_t}^w}{\partial t} \approx \frac{\mathbf{p}_{c_{t+\Delta t}}^w - \mathbf{p}_{c_t}^w}{\Delta t} \\ \mathbf{m}_d(t) &= \frac{\partial \mathbf{p}_{d_t}^w}{\partial t} \approx \frac{\mathbf{p}_{d_{t+\Delta t}}^w - \mathbf{p}_{d_t}^w}{\Delta t}. \end{aligned} \tag{13}$$

According to (10) and (11)

$$\begin{aligned} \mathbf{m}_o &= s \cdot \mathbf{m}_d + \mathbf{m}_c \\ \hat{\mathbf{m}}_o &= \hat{s}\mathbf{m}_d + \mathbf{m}_c = \left(1 + \frac{\Delta s}{s}\right) \mathbf{m}_o - \frac{\Delta s}{s} \mathbf{m}_c. \end{aligned} \tag{14}$$

There are multiple statistical representation methods to measure the correlation relationship. The key objective function formulation is about how to quantify the correlation properly. The basic statistical concept of statistical correlation is cross covariance with time shift τ

$$\begin{aligned} \Sigma_{\hat{o}c}(\tau) &= \mathrm{Cov}(\hat{\mathbf{m}}_o(t), \mathbf{m}_c(t+\tau)) \\ &= \hat{s}\Sigma_{dc}(\tau) + \Sigma_{cc}(\tau) \end{aligned} \tag{15}$$

which is a 3×3 covariance matrix. Particularly, when $\hat{s}$ equals s

$$\Sigma_{oc}(\tau) = s\Sigma_{dc}(\tau) + \Sigma_{cc}(\tau). \tag{16}$$

Additionally

$$\begin{aligned} \Sigma_{\hat{o}\hat{o}}(\tau) &= \mathrm{Cov}(\hat{\mathbf{m}}_o(t), \hat{\mathbf{m}}_o(t+\tau)) \\ &= \hat{s}^2\Sigma_{dd}(\tau) + \hat{s}(\Sigma_{dc}(\tau) + \Sigma_{cd}(\tau)) + \Sigma_{cc}(\tau). \end{aligned} \tag{17}$$

The covariances $\Sigma_{dd}(\tau)$, $\Sigma_{dc}(\tau)$, $\Sigma_{cd}(\tau)$, and $\Sigma_{cc}(\tau)$ can be estimated from the latest N_o observations (The observation number N_o is different from the frame number N of the sliding window in the region-based BA module, and N_o is a large but limited number for bounded computation complexity). More details can also be found in Appendix A.

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VII. Scale Estimation Using Trace Correlation Analysis

In our previous work [11], we utilized a simplified correlation measurement (Frobenius norm) such that a closed-form scale estimation could be quickly derived. However, the corresponding observability condition is too strict to obtain a high estimation rate. It is crucial to select the most reasonable correlation quantification approach since the correlation has a particular physical meaning. In fact, a more robust correlation measurement between two random vectors is trace correlation [39]. In the following, we will first introduce canonical correlation analysis (CCA) and trace correlation, and formulate a more robust objective function for metric scale estimation.

A. CCA and Trace Correlation

Given two random vectors $\mathbf{x} = [x_1, x_2, x_3]^T$ and $\mathbf{y} = [y_1, y_2, y_3]^T$, one may define the cross covariance $\Sigma_{\mathbf{xy}} = \mathrm{Cov}(\mathbf{x}, \mathbf{y})$ to be the 3×3 matrix whose (i, j) entry is the covariance $\mathrm{Cov}(x_i, y_j)$, and $\Sigma_{\mathbf{xx}}$ and $\Sigma_{\mathbf{yy}}$ are the autocovariance matrices. All of them are estimated from the sample data of $\mathbf{x}$ and $\mathbf{y}$. The objective of CCA [40], [41] is to find linear combination vectors of $\mathbf{a}$ and $\mathbf{b}$ such that the correlation $\mathrm{Corr}(\mathbf{a}^T\mathbf{x}, \mathbf{b}^T\mathbf{y})$ is maximized

$$\mathbf{a}_1, \mathbf{b}_1 = \underset{\mathbf{a},\mathbf{b}}{\mathrm{argmax}} \frac{\mathbf{a}^T\Sigma_{\mathbf{xy}}\mathbf{b}}{\sqrt{\mathbf{a}^T\Sigma_{\mathbf{xx}}\mathbf{a}}\sqrt{\mathbf{b}^T\Sigma_{\mathbf{yy}}\mathbf{b}}}. \tag{18}$$

The random variables $\eta = \mathbf{a}_1^T\mathbf{x}$ and $\phi = \mathbf{b}_1^T\mathbf{y}$ are the first pair of canonical variables. The second pair of canonical variables is obtained if the correlation is maximized, with the constraint that the combination vectors are uncorrelated with that of the first pair of canonical variables. This procedure will be continued two more times.

In fact, there exists an inequality

$$\mathrm{Corr}(\mathbf{a}^T\mathbf{x}, \mathbf{b}^T\mathbf{y}) \leq \frac{(\mathbf{a}^T\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}\mathbf{a})^{1/2}}{(\mathbf{a}^T\Sigma_{\mathbf{xx}}\mathbf{a})^{1/2}} \tag{19}$$

and the maximum correlation is obtained when $\mathbf{a}$ is the eigenvector with the maximum eigenvalue of the matrix $\Sigma_{\mathbf{xx}}^{-1}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}$. The subsequent $\mathbf{a}$ are calculated by using eigenvalue decomposition

$$\begin{aligned}\Sigma_{\mathbf{xx}}^{-1}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}} &= Q\Lambda Q^{-1} \\ &= \begin{bmatrix} | & | & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix} \begin{bmatrix} - & \mathbf{a}_1^T & - \\ - & \mathbf{a}_2^T & - \\ - & \mathbf{a}_3^T & - \end{bmatrix}\end{aligned} \tag{20}$$

where $|\lambda_1| \geq |\lambda_2| \geq |\lambda_3|$.

While $\mathbf{b}_i$ are obtained by

$$\mathbf{b}_i \propto \Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}\mathbf{a}_i, i \in \{1, 2, 3\}. \tag{21}$$

The corresponding correlation coefficients between the canonical variables can be represented as

$$\rho_i = \mathrm{Corr}(\mathbf{a}_i^T\mathbf{x}, \mathbf{b_i}^T\mathbf{y}), i \in \{1, 2, 3\}. \tag{22}$$

Trace correlation is defined as

$$\bar{r}_{\mathbf{xy}} = \sqrt{\frac{1}{3}\sum\nolimits_{i=1}^{3}\rho_i^2}. \tag{23}$$

Obviously, the value distribution has a range from 0 to 1, where 0 means strictly uncorrelation of two random vectors and 1 means the two random vectors are strictly correlated. Similar to the correlation coefficient between two random variables, trace correlation is another "normalized" measurement that is not affected by the absolute value, it is transformation invariant because of eigenvalue decomposition. One key property of trace correlation is that, as the name implies, it can be simplified in terms of the trace of a specific matrix

$$\bar{r}_{\mathbf{xy}} = \sqrt{\left(\frac{1}{3}\mathrm{Tr}(\Sigma_{\mathbf{xx}}^{-1}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}})\right)}. \tag{24}$$

The reason is that the eigenvalues of (20) equal to the correlation coefficient squares

$$\lambda_i = \rho_i^2, i \in \{1, 2, 3\} \tag{25}$$

and

$$\sum_{i=1}^{3}\rho_i^2 = \mathrm{Tr}(\Sigma_{\mathbf{xx}}^{-1}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}). \tag{26}$$

The detailed derivation can be found in Appendix B.

B. Objective Function Formulation

On the basis of the definition of trace correlation between two random vectors, the trace correlation between two random processes can be derived. In our formulation, $\mathbf{x}$ and $\mathbf{y}$ correspond to the recovered object motion $\hat{\mathbf{m}}_o$ and the camera motion $\mathbf{m}_c$, respectively. The corresponding trace correlation with time shift τ between the recovered object motion and the camera motion can be represented as

$$\bar{r}_{\hat{o}c}(\tau) = \left(\frac{1}{3}\mathrm{Tr}(\Sigma_{\hat{o}\hat{o}}(0)^{-1}\Sigma_{\hat{o}c}(\tau)\Sigma_{cc}(0)^{-1}\Sigma_{\hat{o}c}(\tau)^T)\right)^{\frac{1}{2}}. \tag{27}$$

The most intuitive method to formulate the objective function is to use the trace correlation of zero time shift, because all the motion data estimated by the VIO and region-based BA using different parts of the same images are synchronized

$$f(\hat{s}) = \bar{r}_{\hat{o}c}(0). \tag{28}$$

However, for a really uncorrelated pair of the recovered object motion and camera motion, the uncorrelation also holds when the time shift changes. Thus, we prefer to formulate the final objective function by considering multiple motion pairs with different time shift values. We formulate the final objective function as

$$f(\hat{s}) = \max_{\tau \in \mathcal{T}} \bar{r}_{\hat{o}c}(\tau) \tag{29}$$

where $\mathcal{T}$ is the set of time shifts. It is actually an upper envelope curve of all the trace correlation curves.

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In the end, the optimal scale can be estimated by minimizing the objective function

$$\hat{s}^* = \arg\min f(\hat{s}). \tag{30}$$

As we know, both $\Sigma_{\hat{o}\hat{o}}(\tau)$ and $\Sigma_{\hat{o}c}(\tau)$ are functions of $\hat{s}$. The expression of trace correlation in terms of $\hat{s}$ is a complicated one. Thus, we solve this minimization problem using scale enumeration in a set S_e ranging from 0 to an upper limit value with a suitable enumeration step.

C. Observability Condition

One key point we want to highlight is that the scale estimator has degenerated cases using a monocular sensor scheme. Take the case that the camera is almost static during tracking for example. Since static camera motion is uncorrelated with arbitrary motion patterns, the objective value is always zero and there will be no optimal scale estimation. In fact, the observability condition is more important than the scale estimation process, as it determines if we can accept the scale estimation or not within a certain observation duration. In addition, our intrinsic scale estimator does not have to be running all the time since what we estimate is the intrinsic scale of the tracked object. Estimating the intrinsic scale once in a while when the observability condition is satisfied is totally acceptable because of this property.

Thanks to the transformation invariability property and "normalized" measurement feature of trace correlation, this correlation evaluation is more robust and accurate than that of using the Frobenius norm [11]. The corresponding observability condition can be softened, resulting in a higher scale estimation rate and more accurate scale estimation.

Recall (14). One degenerated case is that $\mathbf{m}_d$ and $\mathbf{m}_c$ are collinear

$$\mathbf{m}_d = \alpha \mathbf{m}_c. \tag{31}$$

This seldom happens in a real-world tracking process, but when it happens, an inaccurate scale estimate ($\alpha\hat{s} + 1 = 0$) leads to zero $\hat{\mathbf{m}}_o$ with zero $\Sigma_{\hat{o}c}$, which is the optimal result of minimizing the objective function. If $\mathbf{m}_d$ and $\mathbf{m}_c$ are collinear, the trace correlation $\bar{r}_{dc}$ between $\mathbf{m}_d$ and $\mathbf{m}_c$ is 1 since they are strictly correlated

$$\bar{r}_{dc} = \left(\frac{1}{3}\mathrm{Tr}(\Sigma_{dd}^{-1}\Sigma_{dc}\Sigma_{cc}^{-1}\Sigma_{cd})\right)^{\frac{1}{2}}. \tag{32}$$

We obtain the first condition

$$\bar{r}_{dc} < 1. \tag{33}$$

Another degenerated case is that $\mathbf{m}_d$ and $\mathbf{m}_c$ are uncorrelated. Recall (15). If $\Sigma_{dc} = \mathbf{0}$, $\Sigma_{\hat{o}c}$ always equals Σ_{cc}, and the optimal scale can never be recovered by minimizing the objective function. The second condition can be represented as

$$\bar{r}_{dc} > 0. \tag{34}$$

In addition, since all the optimization is based on the camera motion and the camera motion itself are strictly correlated, the object motion and the camera motion are weakly correlated. Generally, only static motion has no strict correlation with itself, and we simply check the trace of Σ_{cc}^2, which means the sum of eigenvalue squares of the matrix Σ_{cc}. As for the weak correlation check between the object motion and the camera motion, we check the recovered object motion and the camera motion, namely the optimal objective function value, instead. It is actually a necessary condition rather than a sufficient condition. We represent these two conditions as

$$\begin{aligned} \mathrm{Tr}(\Sigma_{cc}^2) &> 0 \\ f(\hat{s}^*) &\to 0. \end{aligned} \tag{35}$$

In practice, we tighten the first three conditions for more robust estimation, whereas for the last condition, we find that some correlation between the camera motion and object motion is acceptable because of the use of trace correlation. So we lighten the last condition to some extent. Finally, we merge the first two conditions as one, and summarize the whole observability condition as follows:

$$\begin{aligned} \rho_b < \mathrm{Tr}(\Sigma_{dd}^{-1}\Sigma_{dc}\Sigma_{cc}^{-1}\Sigma_{cd}) < \rho_u \\ \mathrm{Tr}(\Sigma_{cc}^2) > \zeta_b \\ f(\hat{s}^*) < \epsilon_u \end{aligned} \tag{36}$$

where ρ_b and ρ_u are the bottom threshold and upper threshold for the trace correlation between $\mathbf{m}_d$ and $\mathbf{m}_c$, ζ_b is the bottom threshold for autocovariance of camera motion, and ϵ_u is the upper threshold for the objective function. All the thresholds are set empirically as a tradeoff between the estimation accuracy and the acceptance rate, which will be detailed in Section X for implementation. Compared with the observability condition of that using the Frobenius norm [11], the scale estimation rate is significantly increased since the key checking condition of the objective function is softened.

The whole algorithm is summarized as Algorithm 1.

VIII. System Limitation

A severe degenerated case of the region-based BA module is a phenomenon called affine projection degeneration from normal perspective projection [33]. It is caused by the perspective effect of the object projected on the image plane being overwhelmed by the measurement noise, especially when the depth variation of the object is relatively small compared with its depth to the camera. It seldom happens in traditional visual BA methods, where whole images are used as input, since the depth variation of the whole image scene is large enough. However, it may happen in our region-based BA method, where only object regions are used as input, because the object depth variation can be relatively small compared with the object depth. When this degenerated case happens, the object structure cannot be reconstructed accurately and the estimated camera poses in the object frame will be incorrect. Therefore, the tracked object cannot be too far away from the camera, with the specific working range depending on the size of the object, and the 2-D object region extracted from the raw image cannot be too small. We also note that at least 50 features should be extracted from the object region as we use a feature-based tracking frontend in the region-based

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Algorithm 1: Metric Scale Estimation Algorithm.

Input: camera poses in world frame $\mathbf{p}_{c_t}^w, \mathbf{R}_{c_t}^w$, up-to-scale object positions in camera frame $\bar{\mathbf{p}}_{o_t}^{c_t}$

Output: metric scale $\hat{s}$ ($\hat{\mathbf{p}}_{o_t}^{c_t} = \hat{s}\bar{\mathbf{p}}_{o_t}^{c_t}$)

$\mathbf{p}_{c_t}^w, \mathbf{p}_{d_t}^w = \mathbf{R}_{c_t}^w \bar{\mathbf{p}}_{o_t}^{c_t}$

$\mathbf{p}_{c_t}^w \rightarrow \mathbf{m}_c, \mathbf{p}_{d_t}^w \rightarrow \mathbf{m}_d$ ▷ Extract motion features

if not $\rho_b < \mathrm{Tr}(\Sigma_{dd}^{-1}\Sigma_{dc}\Sigma_{cc}^{-1}\Sigma_{cd}) < \rho_u$ **then**

  return

end if ▷ First observability subcondition

if $\mathrm{Tr}(\Sigma_{cc}^2) \leq \zeta_b$ **then**

  return

end if ▷ Second observability subcondition

for $\hat{s} \in S_e$ **do** ▷ Scale enumeration

  for $\tau \in \mathcal{T}$ **do** ▷ Time shift enumeration

    $\Sigma_{dd}(\tau), \Sigma_{dc}(\tau), \Sigma_{cd}(\tau), \Sigma_{cc}(\tau)$ ▷ Update motion statistics

    $\Sigma_{\hat{o}c}(\tau) = \hat{s}\Sigma_{dc}(\tau) + \Sigma_{cc}(\tau)$

    $\Sigma_{\hat{o}\hat{o}}(\tau) = \hat{s}^2\Sigma_{dd}(\tau) + \hat{s}(\Sigma_{dc}(\tau) + \Sigma_{cd}(\tau)) + \Sigma_{cc}(\tau)$

    $\bar{r}_{\hat{o}c}(\tau) = (\frac{1}{3}\mathrm{Tr}(\Sigma_{\hat{o}\hat{o}}(0)^{-1}\Sigma_{\hat{o}c}(\tau)\Sigma_{cc}(0)^{-1}\Sigma_{\hat{o}c}(\tau)^T))^{\frac{1}{2}}$

  end for

end for

$f(\hat{s}) = \mathbf{max}_\tau \bar{r}_{\hat{o}c}(\tau)$ ▷ Upper envelope curve

$s^* = \underset{\hat{s}}{\mathrm{argmin}}\, f(\hat{s})$

if $f(\hat{s}^*) < \epsilon_u$ **then** ▷ Third observability subcondition

  return $\hat{s}*$ ▷ Accept estimation

else

  return

end if

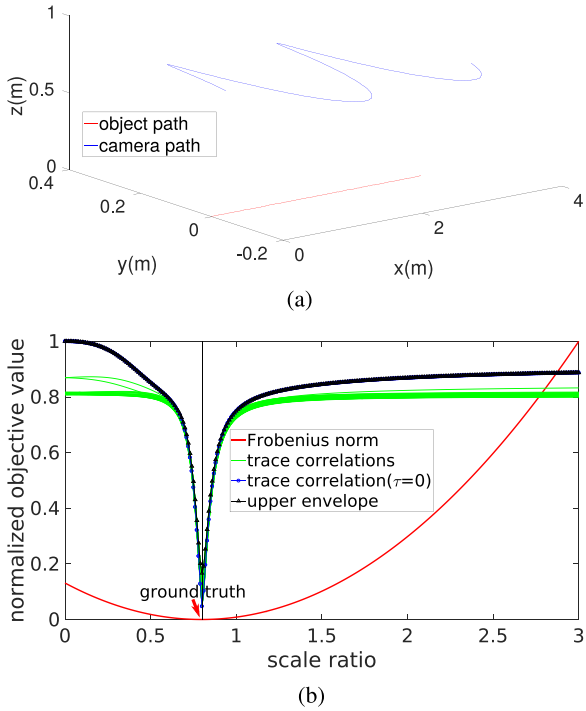


Fig. 6. Simulation Trial 1. All the objective formulation methods can obtain unbiased scale estimation in this case. In (b) the ground truth scale ratio is denoted by a vertical line segment. The Frobenius norm [11] is denoted as a red curve, the trace correlations with different time shift values are denoted as green curves, and the trace correlation with time shift $\tau = 0$ ((28)) is denoted as a blue curve marked with circles. The upper envelope curve of all the trace correlation curves ((29)) is denoted as a black curve marked with triangles. (a) Path. (b) Objective function.

BA. To alleviate this degenerated case as much as possible, we use a high-resolution camera instead.

For the metric scale estimation module, we have discussed the possible degenerated cases in Section VII-C. Although we derive the corresponding observability condition to avoid inaccurate scale estimations caused by the degenerated cases, successful metric scale recovery is conditional. Fortunately, estimating the object scale only when the observability condition is satisfied is acceptable because the intrinsic scale of a certain object is time invariant.

IX. Simulation

In order to evaluate different objective function formulation methods of scale estimation, we build a simulation environment for analysis. In addition, since it is difficult to realize some degenerated cases in real experiments, we test the degenerated cases in the same simulation environment rather than in the real experiments. The simulation is designed to be focused on the metric scale estimation part, in which none of the perception errors of the VIO and region-based BA are considered. To be specific, we first construct the camera trajectory and object trajectory in the world frame, respectively, and directly use the camera trajectory as the VIO result. For the object perception, we generate the up-to-scale object observation in the camera frame by projecting the 3-D object trajectory to the 2-D image plane. In this way, the metric scale information of the object observation is discarded. White noise is also added to both the VIO result and object perception. The target is to recover the metric scale of the object using the simulation perception data. Finally, we test different objective formulation methods and study the degenerated cases by plotting the objective values regarding the sample enumeration scale ratios, ranging from 0 to 3 with step 0.005.

A. Comparison of Objective Formulation Methods

We discuss two typical motion combinations.

1) The object has straight line motion with constant velocity (inactive motion), while the camera has sinusoidal motion.
2) Both the object and the camera have random motions in 3-D space, and the camera motion and object motion are correlated to some extent in all three directions $(\mathrm{x}, \mathrm{y}, \mathrm{z})$.

We test multiple motion combinations for each case by generating random object trajectories and camera trajectories, respectively. The patterns of the plotted objective function curves are similar, typical results of the object and camera trajectory, objective curves of different formulation methods are plotted in Figs. 6 and 7. In Figs. 6(b) and 7(b), the ground truth scale

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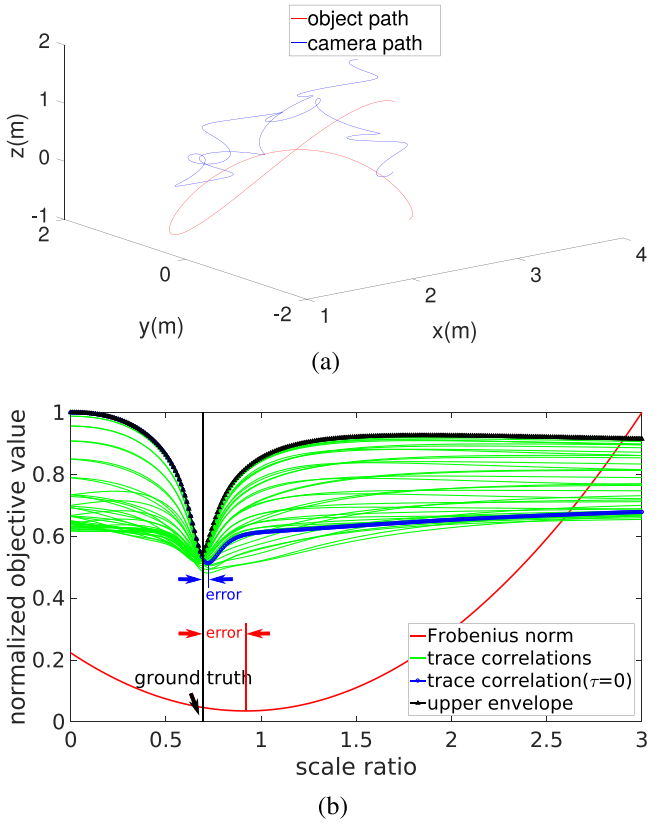


Fig. 7. Simulation Trial 2. The Frobenius norm fails to obtain accurate scale estimation (red error), the trace correlation with 0 time shift has small estimation bias in this case (blue error). The upper envelope curve of all the trace correlation curves has the best performance. (a) Path. (b) Objective function.

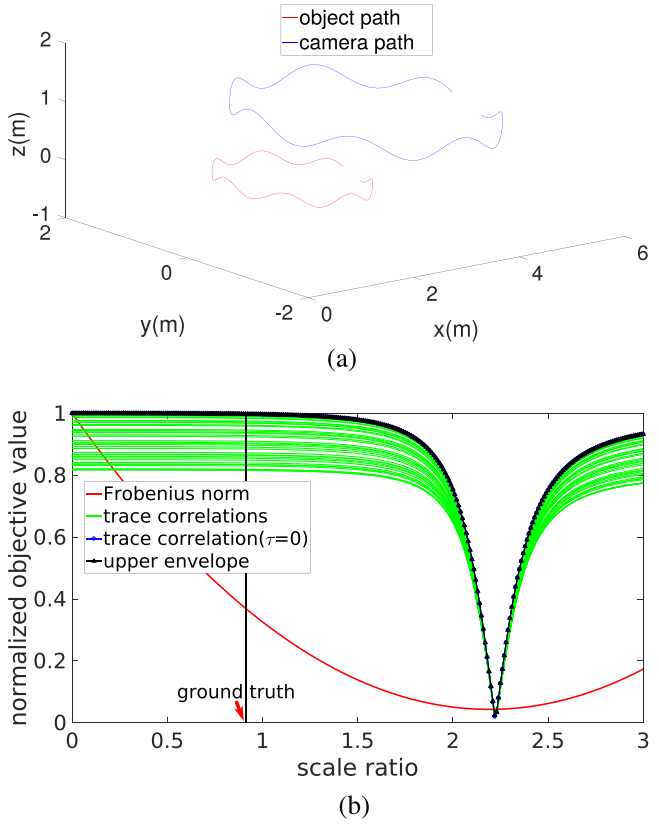


Fig. 8. Degenerated case 1: The camera motion ($\mathbf{m}_c$) and the motion difference ($\mathbf{m}_d$) are collinear ($\bar{r}_{dc} = 1$). (a) Path. (b) Objective function.

ratio is denoted by a vertical line segment. The Frobenius norm [11] is denoted as a red curve, the trace correlations with different time shift values are denoted as green curves, in which the one with time shift $\tau = 0$ (28) is denoted as a blue curve marked with circles. The upper envelope curve of all the trace correlation curves (29) is denoted as a black curve marked with triangles. According to (30), the optimal point with the minimum objective value is regarded as the scale estimation result of that objective formulation method. The estimation errors are depicted in the figures. In Fig. 6, we can see that for the strict uncorrelated motion case (Trial 1), all the objective formulation methods lead to accurate scale estimations, while in Fig. 7, for the general motion case (Trial 2), the estimation errors are also depicted, and it is shown the trace correlation with 0 time shift has relatively better discrimination ability than those with other time shift values. However, only the upper envelope curve of all the trace correlation curves has an unbiased estimation.

B. Degenerated Cases

Consider four degenerated cases introduced in Section VII-C:

1) The camera motion ($\mathbf{m}_c$) and the motion difference ($\mathbf{m}_d$) are collinear ($\bar{r}_{dc} = 1$).
2) The camera motion ($\mathbf{m}_c$) and the motion difference ($\mathbf{m}_d$) are uncorrelated ($\bar{r}_{dc} = 0$).
3) The camera motion ($\mathbf{m}_c$) is static ($\mathrm{Tr}(\Sigma_{cc}^2) = 0$).
4) The camera motion ($\mathbf{m}_c$) and the object motion ($\mathbf{m}_o$) are strictly correlated ($\bar{r}_{oc} = 1$).

The object and camera trajectory, objective curves of different formulation methods for the four degenerated cases are plotted respectively from Figs. 8 to 11, where the true scale is denoted by a vertical line segment. For the degenerated cases, all the formulations fail to recover the real object scale. However, these cases can be detected and dropped by checking the observability condition (see Section VII-C).

X. Experimental Results

A. Implementation Details

Our minimal sensor suite consists of an mvBlueFOX-MLC202bG grayscale camera with a wide-angle lens that captures 1280×960 images at 24 Hz and a Microstrain 3-DM-GX4 IMU that runs at 500 Hz. Both VINS [26] and the 2-D tracker work on downsampled images of 640×480 resolution. We adopt YOLO [36] as the 2-D detector for 2-D tracking initialization, and use Re-3 [32] as the 2-D tracker in our implementation. The frame number of the sliding window in the region-based BA module is 15 ($N = 15$). The number of motion observations we use for metric scale estimation is 200 ($N_o = 200$). The extracted motion feature is the velocity ($k = 1$ in (12)). The scale enumeration range S_e is from 0 to 3 with step 0.005

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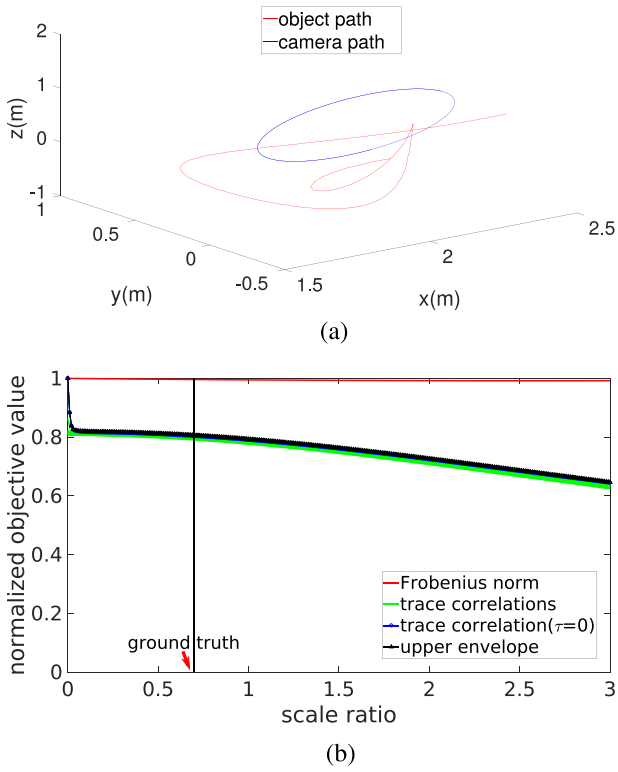


Fig. 9. Degenerated case 2: The camera motion ($\mathbf{m}_c$) and the motion difference ($\mathbf{m}_d$) are uncorrelated ($\bar{r}_{dc} \approx 0$). (a) Path. (b) Objective function.

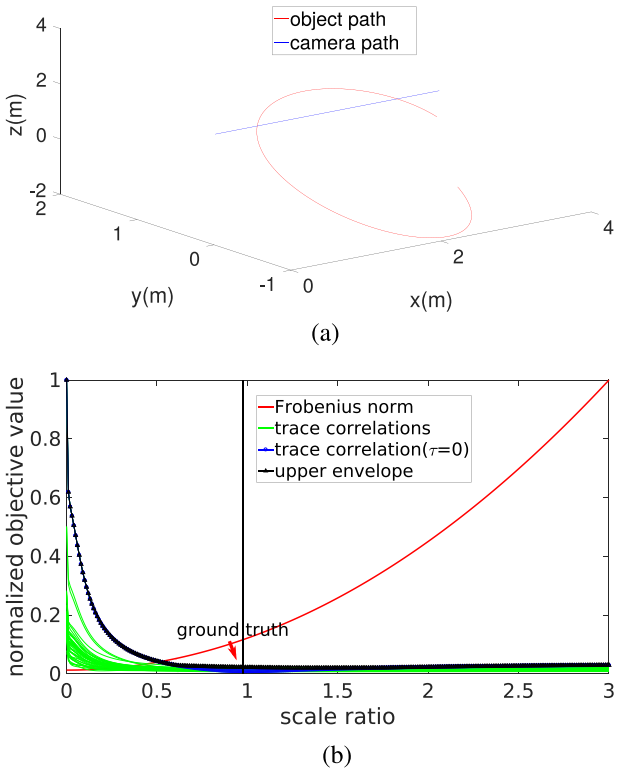


Fig. 10. Degenerated case 3: The camera motion ($\mathbf{m}_c$) is static ($\mathrm{Tr}(\Sigma_{cc}^2) = 0$). (a) Path. (b) Objective function.

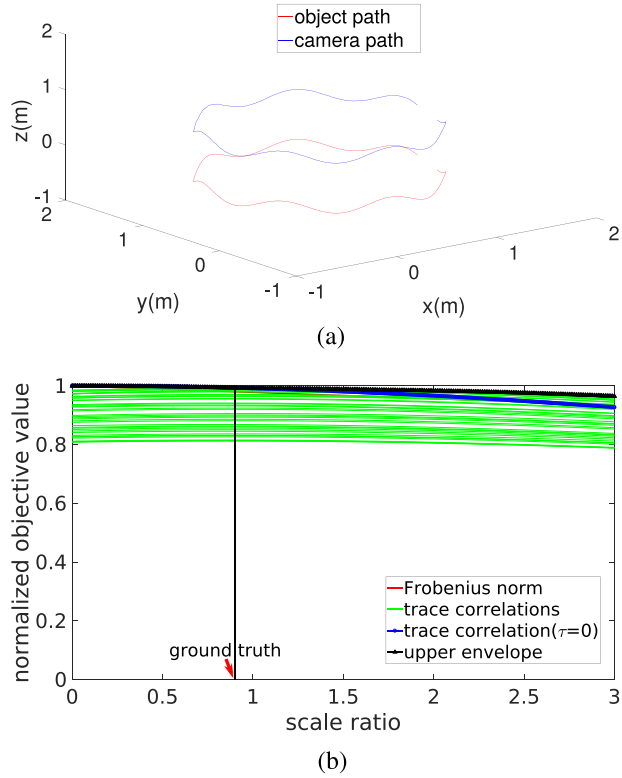


Fig. 11. Degenerated case 4: The camera motion ($\mathbf{m}_c$) and the object motion ($\mathbf{m}_o$) are strictly correlated ($\bar{r}_{oc} = 1$). (a) Path. (b) Objective function.

for the handheld tracking case in an indoor environment. The time shift enumeration range $\mathcal{T} = \{-20\Delta t, \ldots, 0, \ldots, 20\Delta t\}$ ($\Delta t = 1/24$ s). The threshold values used for observability condition check are $\rho_b = 0.3$, $\rho_u = 0.78$, $\zeta_b = 1\mathrm{e}{-4} m^4/s^4$, and $\epsilon_u = 0.55$. The whole system runs real time on a desktop computer with an i7-8700K CPU and a GTX1080Ti GPU, in which the GPU is only for the learning-based 2-D detector and tracker.

B. Accuracy of 6-DoF Object Pose Estimation

The goal of the experiments is to validate the recovered object pose in the world frame ($\mathbf{R}_o^w$ and $\mathbf{p}_o^w$) by involving the camera pose in the world frame and the estimated metric scale. Consider that no similar problem formulations have been shown in relevant works: model-based and learning-based methods inherently encode the object scale and recover the metric object pose with only one image, whereas our method needs to accumulate a period of camera and object motion observations for object scale estimation and metric object pose reconstruction. Thus, we evaluate our object tracking method with a comparison to a motion capture system (OptiTrack) as the ground truth. The estimated pose is aligned to the ground truth pose with all the estimates for validation. We use Z-Y-X Euler angles for 3-D rotation representation, so yaw, pitch, roll errors together denote the rotation estimation error. We select several rigid objects as the tracking target, including a remote-controlled model car, a toy robot, etc.

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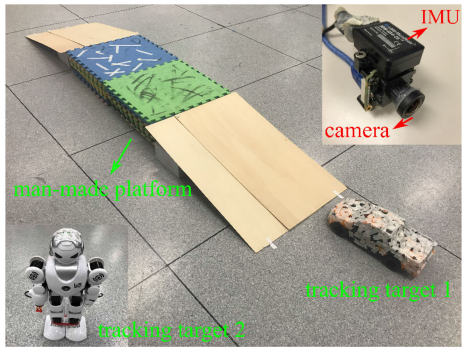


Fig. 12. Minimal sensor suite and sample tracking targets. The experimental environment and the man-made platform for the car are also shown. The tape attached to the objects is used to cover the excessively shiny parts that may deactivate the motion capture system for ground truth collection.

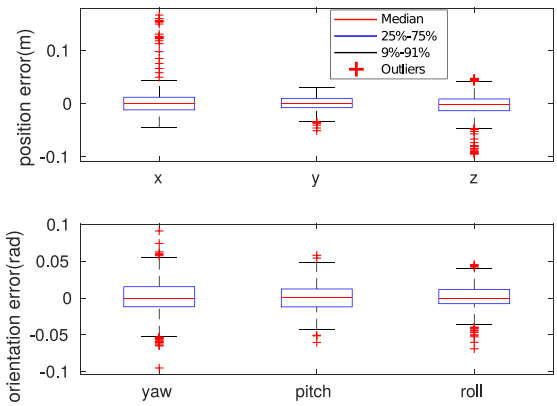


Fig. 13. Boxplot of the camera position and orientation estimation errors compared with ground truth of VINS [26] using our sensor suite.

Both the minimal sensor suite and sample tracking targets are shown in Fig. 12. For example, the car is controlled to perform multiple arbitrary motion patterns and styles, including linear motion, turning motion, climbing motion, descending motion, aggressive motion, etc.

Given the estimated metric scale with the observability condition satisfied, we can recover the object poses in the world frame according to (2). It is evident that the final object pose estimation relies on the VIO ($\hat{\mathbf{p}}^w_{c_t}$ and $\hat{\mathbf{R}}^w_{c_t}$), up-to-scale 3-D tracking ($\hat{\bar{\mathbf{p}}}^{c_t}_{o_t}$ and $\hat{\mathbf{R}}^{c_t}_{o_t}$), and metric scale estimation results ($\hat{s}$). Obviously the metric scale estimator only affects the position estimation results. We first measure the estimation error of VIO for our sensor suite. The position and orientation errors of camera pose is shown in Fig. 13 with a standard deviation of $\{0.0214, 0.0188, 0.0150\}$ (rad) in yaw, pitch, and roll, and a standard deviation of $\{0.0230, 0.0128, 0.0208\}$ (m) in the x, y, and z positions.

Finally, the object position and orientation errors compared with ground truth of tracking the remote-controlled car is shown in Fig. 14, with a standard deviation of $\{0.0698, 0.0288, 0.0305\}$ (rad) in yaw, pitch, and roll, and a standard deviation of $\{0.0897, 0.0701, 0.0538\}$ (m) in the x, y, and z of our method. The estimation error of our previous work [11] is also illustrated for comparison. Fig. 15 shows the case of tracking the toy

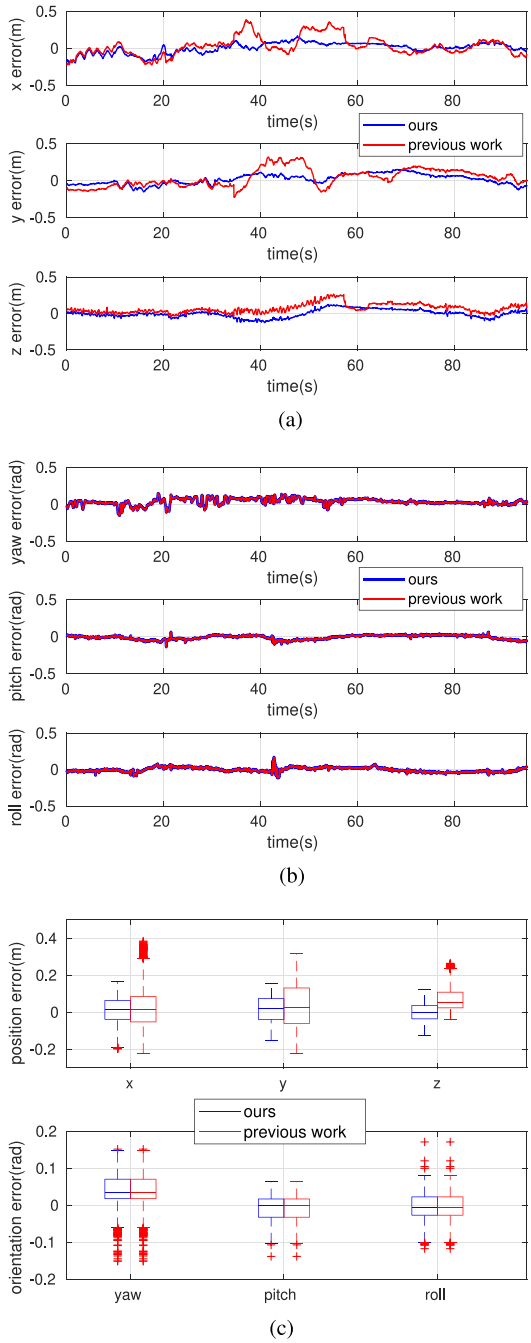


Fig. 14. Object position and orientation estimation error compared with ground truth of Trial 1 (tracking the remote-controlled car). The position estimation of the proposed method is more accurate than our previous work [11]. The orientation estimations are identical because they are completely decided by the region-based BA module. (a) Position error. (b) Orientation error. (c) Boxplot of error statistics.

robot. Generally, the experiment results show that the estimation accuracy of our 6-DoF object tracking system is acceptable, in which the position estimation based on trace correlation is better than that of our previous work. Timing statistics of each module are given in Table I. The metric scale estimation is accepted only when the corresponding observability condition is satisfied. It can be seen that the acceptance rate of the estimated metric

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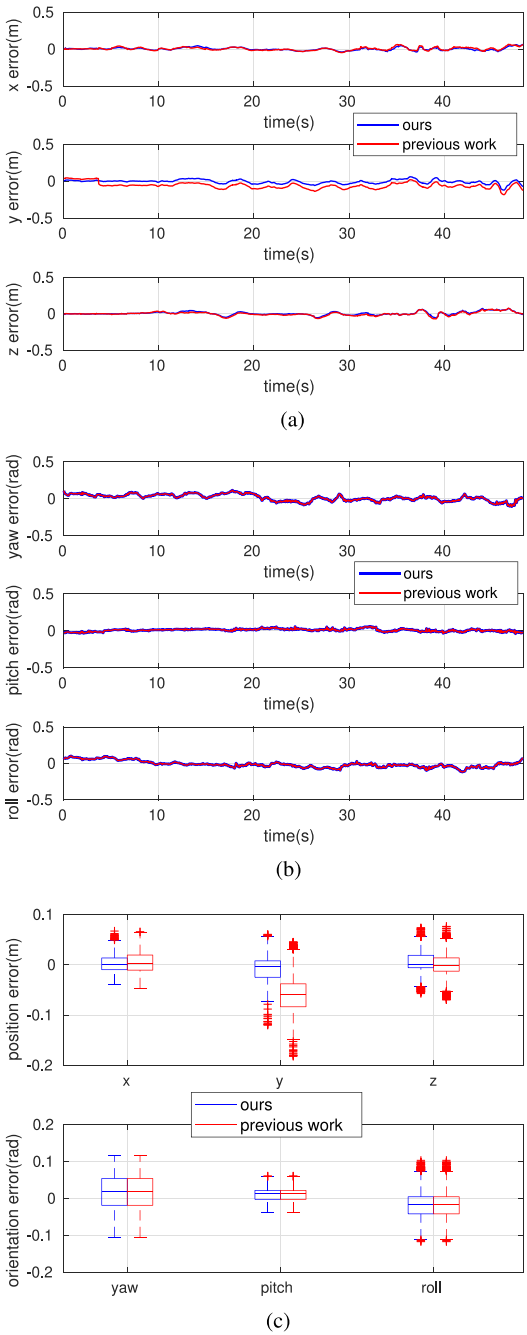


Fig. 15. Object position and orientation estimation error compared with ground truth of Trial 2 (tracking the toy robot). The position estimation of the proposed method is more accurate than our previous work [11]. The orientation estimations are identical because they are completely decided by the region-based BA module. (a) Position error. (b) Orientation error. (c) Boxplot of error statistics.

scale introduced in this paper is much higher than in our previous work, as discussed in Section VII-C.

To vividly present the 3-D tracking performance, we demonstrate an AR effect on the tracked object. Actually, we render virtual 3-D coordinates (each axis = 15 cm) on the object surface according to the estimated object pose with respect to the camera frame. Several screenshots of tracking the car are shown in Fig. 16, where the object coordinates are rendered according

TABLE I
TIMING STATISTICS

Modules	Time (ms)	Rate (Hz)	Acceptance rate (Hz)*
2D detector	40	24	-
2D tracker	15	24	-
VINS [25]	35	24	-
region-based BA	30	24	-
scale estimator	5	24	6
previous estimator [11]	3	24	0.1

*Acceptance rate is only available for metric scale estimator. It measures the rate when the observability condition is satisfied (see Section VII-C).

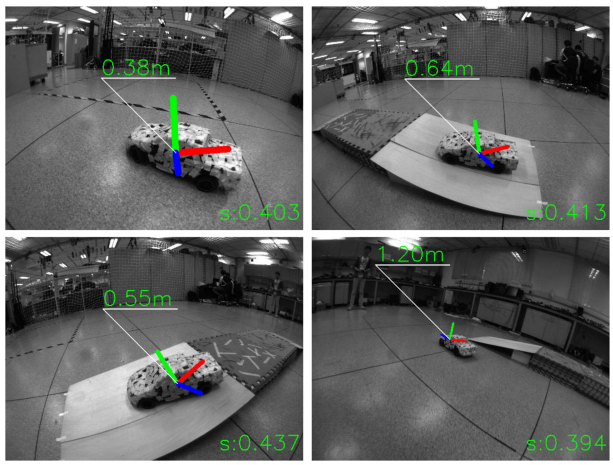


Fig. 16. AR effect based on 3-D tracking results. The metric 6-DoF pose of the tracked object is estimated according to the region-based BA with metric scale estimation. The corresponding object coordinates are rendered according to the tracking results to vividly represent the object pose. The estimated scale ratio s is shown at the bottom of each image, and the distance between the object coordinates' principle and the camera center is also demonstrated.

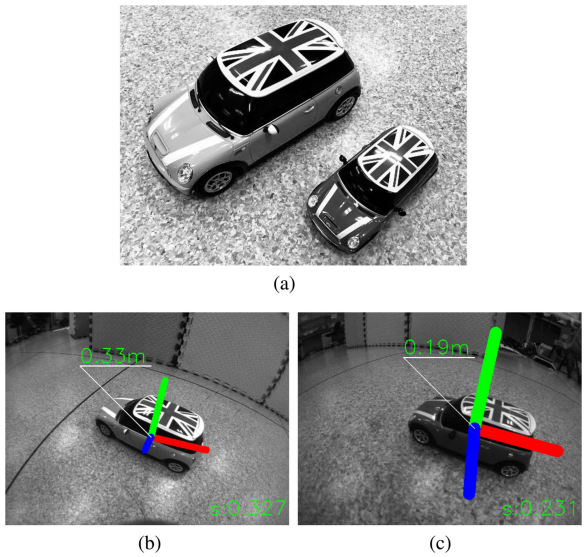


Fig. 17. Tracking comparison between a large object (1 : 14 real car) and a small object (1 : 24 real car) with a similar appearance. The virtual object coordinates (each axis = 15 cm) are rendered according to the tracking results. The estimated scale ratio s is shown at the bottom of each image and the distance between the coordinates' principle and the camera center is also demonstrated. The car sizes can be distinguished by the relative sizes of the coordinates in the images. (a) Objects for scale ambiguity test. (b) Large object (1 : 14 real car). (c) Small object (1 : 24 real car).

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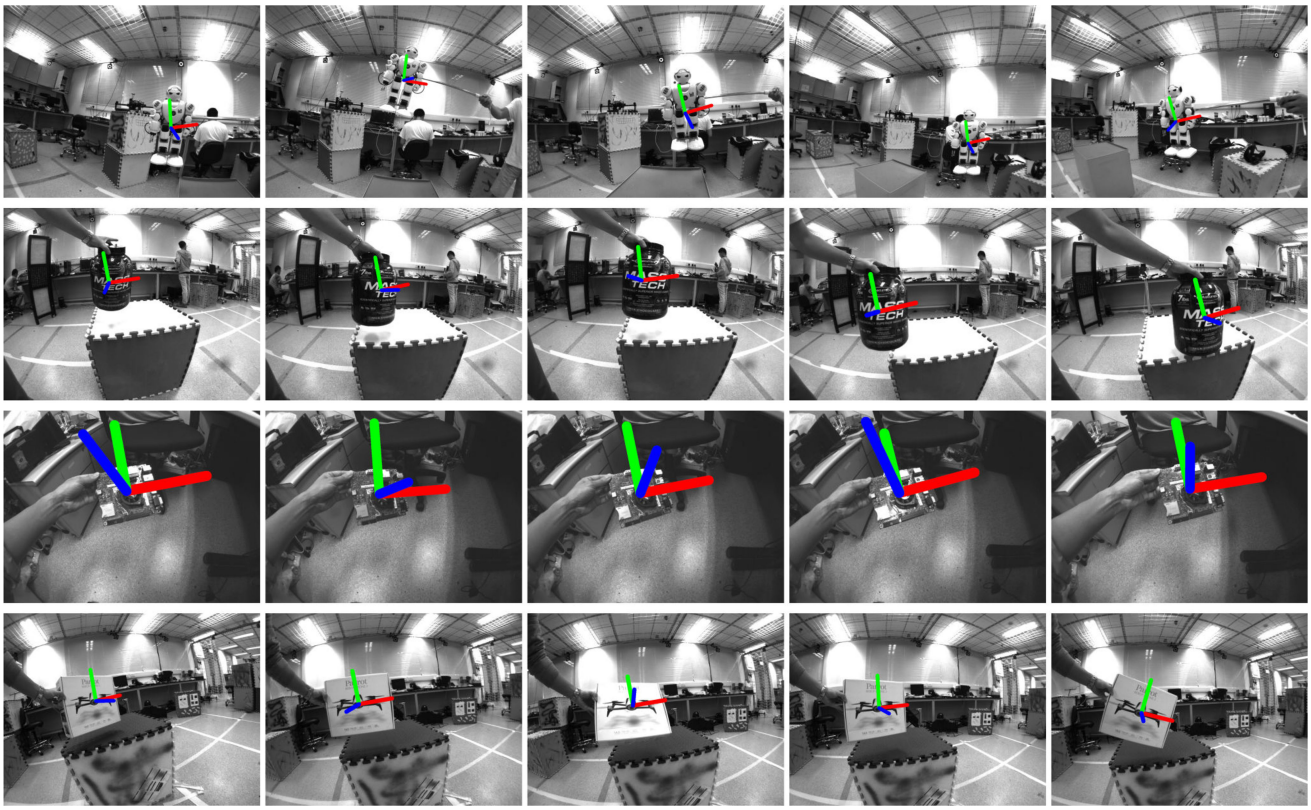


Fig. 18. Tracking results of general objects. The metric 6-DoF pose of the tracked object is obtained by our 3-D tracking system, and the corresponding object coordinate (each axis = 15 cm) is rendered in the image to vividly represent the tracking result. First row: a toy robot. Second row: a plastic bottle. Third row: a development board. Last row: a packing box.

to the 3-D tracking results such that we can intuitively judge the tracking performance by monitoring the position and orientation change of the object coordinates. The rendered coordinates can be conveniently replaced by other 3-D models for specific AR applications. Sample tracking results of more general objects with AR effect are shown in Fig. 18, and further tracking details are shown in the attached video.

C. Scale Ambiguity Test

One important advantage of our geometry-based metric scale estimator is that it can estimate the real intrinsic scale of the tracked object, which means we can track and distinguish two objects that have a similar appearance but different sizes. For the model-based or learning-based methods using a monocular camera, since both of them have a scale prior of a specific object, both will be confused by scale ambiguity, resulting in inaccurate scale estimations.

Instead of using a real car and the corresponding toy model car, we use another two toy model cars with a similar appearance but different sizes for comparison. We judge the scale awareness ability through the rendered virtual coordinates that have fixed size (each axis = 15 cm). The 3-D tracking comparison is shown in Fig. 17, where the coordinates are relatively small for the large car and relatively large for the small car. Thanks to the real scale awareness ability, our system can measure the metric scale of the tracked object using monocular visual-inertial sensing.

XI. Conclusion

In this paper, we proposed a modular 3-D tracking system for 6-DoF pose estimation of dynamic objects with metric scale recovery capability using monocular visual-inertial sensing. The tracking system can handle generally rigid objects based on motion correlation analysis, while no scale prior of the tracked object is needed. We proved the feasibility of the proposed 3-D tracking system with observability condition check, which can avoid multiple unobservable or weakly observable situations. Experimental results validate our idea, showing acceptable estimation accuracy with the proposed method, while an AR demonstration presents a potential application scenario.

Thanks to the modular design of our system, each module can be conveniently replaced for better overall performance. For example, the corner feature association method in the region-based BA module can be replaced by edge-based and dense methods to handle featureless objects. A smart 2-D object region generation module with object indexing and recognition abilities can handle multiple object and similar object tracking problems. A smart region-based BA module with robust relocalization ability can handle temporal object tracking loss in the region

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generation module. In addition, the implementation of the 3-D tracking system on mobile devices is one direction for future work.

APPENDIX A
MOTION STATISTICS

According to the definition of cross covariance, the one between the recovered object motion and camera motion can be derived as

$$\begin{aligned}\Sigma_{\hat{o}c} &= \mathrm{Cov}(\hat{\mathbf{m}}_o, \mathbf{m}_c) \\ &= E[(\hat{\mathbf{m}}_o - E[\hat{\mathbf{m}}_o])(\mathbf{m}_c - E[\mathbf{m}_c])^T] \\ &= E[(\hat{s}\mathbf{m}_d + \mathbf{m}_c - E[\hat{s}\mathbf{m}_d + \mathbf{m}_c])(\mathbf{m}_c - E[\mathbf{m}_c])^T] \\ &= \hat{s}\mathrm{Cov}(\mathbf{m}_d, \mathbf{m}_c) + \mathrm{Cov}(\mathbf{m}_c, \mathbf{m}_c) \\ &= \hat{s}\Sigma_{dc} + \Sigma_{cc}.\end{aligned} \tag{37}$$

The autocovariance of the recovered object motion can be represented as

$$\begin{aligned}\Sigma_{\hat{o}\hat{o}} &= \mathrm{Cov}(\hat{\mathbf{m}}_o, \hat{\mathbf{m}}_o) \\ &= E[(\hat{\mathbf{m}}_o - E[\hat{\mathbf{m}}_o])(\hat{\mathbf{m}}_o - E[\hat{\mathbf{m}}_o])^T] \\ &= E[(\hat{s}\mathbf{m}_d + \mathbf{m}_c - E[\hat{s}\mathbf{m}_d + \mathbf{m}_c])(\hat{\mathbf{m}}_o - E[\hat{\mathbf{m}}_o])^T] \\ &= \hat{s}^2\mathrm{Cov}(\mathbf{m}_d, \mathbf{m}_d) + \hat{s}\mathrm{Cov}(\mathbf{m}_d, \mathbf{m}_c) \\ &\quad + \hat{s}\mathrm{Cov}(\mathbf{m}_c, \mathbf{m}_d) + \mathrm{Cov}(\mathbf{m}_c, \mathbf{m}_c) \\ &= \hat{s}^2\Sigma_{dd} + \hat{s}(\Sigma_{dc} + \Sigma_{cd}) + \Sigma_{cc}.\end{aligned} \tag{38}$$

Both of these are functions of the estimated scale $\hat{s}$.

The covariance between two random processes depends on the time shift τ. Both the autocovariance and cross-covariance matrices can be estimated from N_o observations

$$\begin{aligned}&\Sigma_{dd}(\tau) \approx \\ &\frac{\sum_{n=0}^{N_o-1}(\mathbf{m}_d(n\Delta t) - \hat{\mathbf{m}}_d(0))(\mathbf{m}_d(n\Delta t + \tau) - \hat{\mathbf{m}}_d(\tau))^T}{N_o - 1} \\ &\Sigma_{dc}(\tau) \approx \\ &\frac{\sum_{n=0}^{N_o-1}(\mathbf{m}_d(n\Delta t) - \hat{\mathbf{m}}_d(0))(\mathbf{m}_c(n\Delta t + \tau) - \hat{\mathbf{m}}_c(\tau))^T}{N_o - 1} \\ &\Sigma_{cd}(\tau) \approx \\ &\frac{\sum_{n=0}^{N_o-1}(\mathbf{m}_c(n\Delta t) - \hat{\mathbf{m}}_c(0))(\mathbf{m}_d(n\Delta t + \tau) - \hat{\mathbf{m}}_d(\tau))^T}{N_o - 1} \\ &\Sigma_{cc}(\tau) \approx \\ &\frac{\sum_{n=0}^{N_o-1}(\mathbf{m}_c(n\Delta t) - \hat{\mathbf{m}}_c(0))(\mathbf{m}_c(n\Delta t + \tau) - \hat{\mathbf{m}}_c(\tau))^T}{N_o - 1}\end{aligned} \tag{39}$$

where

$$\begin{aligned}&\hat{\mathbf{m}}_d(0) = \\ &\frac{1}{N_o}\sum_{n=0}^{N_o-1}\mathbf{m}_d(n\Delta t),\ \hat{\mathbf{m}}_d(\tau) = \frac{1}{N_o}\sum_{n=0}^{N_o-1}\mathbf{m}_d(n\Delta t + \tau) \\ &\hat{\mathbf{m}}_c(0) = \\ &\frac{1}{N_o}\sum_{n=0}^{N_o-1}\mathbf{m}_c(n\Delta t),\ \hat{\mathbf{m}}_c(\tau) = \frac{1}{N_o}\sum_{n=0}^{N_o-1}\mathbf{m}_c(n\Delta t + \tau).\end{aligned} \tag{40}$$

APPENDIX B
TRACE CORRELATION

Define

$$\mathbf{c} = \Sigma_{\mathbf{xx}}^{1/2}\mathbf{a}, \mathbf{d} = \Sigma_{\mathbf{yy}}^{1/2}\mathbf{b} \tag{41}$$

so we have

$$\mathrm{Corr}(\mathbf{a}^T\mathbf{x}, \mathbf{b}^T\mathbf{y}) = \frac{\mathbf{c}^T\Sigma_{\mathbf{xx}}^{-1/2}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1/2}\mathbf{d}}{\sqrt{\mathbf{c}^T\mathbf{c}}\sqrt{\mathbf{d}^T\mathbf{d}}}. \tag{42}$$

By the Cauchy–Schwarz inequality, we have

$$\begin{aligned}&(\mathbf{c}^T\Sigma_{\mathbf{xx}}^{-1/2}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1/2})\mathbf{d} \leq \\ &(\mathbf{c}^T\Sigma_{\mathbf{xx}}^{-1/2}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1/2}\Sigma_{\mathbf{yy}}^{-1/2}\Sigma_{\mathbf{yx}}\Sigma_{\mathbf{xx}}^{-1/2}\mathbf{c})^{1/2}(\mathbf{d}^T\mathbf{d})^{1/2}.\end{aligned} \tag{43}$$

Thus

$$\mathrm{Corr}(\mathbf{a}^T\mathbf{x}, \mathbf{b}^T\mathbf{y}) \leq \frac{(\mathbf{c}^T\Sigma_{\mathbf{xx}}^{-1/2}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}\Sigma_{\mathbf{xx}}^{-1/2}\mathbf{c})^{1/2}}{(\mathbf{c}^T\mathbf{c})^{1/2}} \tag{44}$$

where $\mathbf{c} = \Sigma_{\mathbf{xx}}^{1/2}\mathbf{a}$, and the equality holds if the vectors $\mathbf{d}$ and $\Sigma_{\mathbf{yy}}^{-1/2}\Sigma_{\mathbf{yx}}\Sigma_{\mathbf{xx}}^{-1/2}\mathbf{c}$ are collinear. The maximum of the correlation is attained if $\mathbf{c}$ is the eigenvector with the maximum eigenvalue of the matrix $\Sigma_{\mathbf{xx}}^{-1/2}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}\Sigma_{\mathbf{xx}}^{-1/2}$ by the Rayleigh quotient [42]. Thus

$$\begin{aligned}\Sigma_{\mathbf{xx}}^{-1/2}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}\Sigma_{\mathbf{xx}}^{-1/2}\mathbf{c}_1 &= \lambda_1\mathbf{c}_1 \\ \Sigma_{\mathbf{xx}}^{-1/2}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}\Sigma_{\mathbf{xx}}^{-1/2}\Sigma_{\mathbf{xx}}^{1/2}\mathbf{a}_1 &= \lambda_1\Sigma_{\mathbf{xx}}^{1/2}\mathbf{a}_1 \\ \Sigma_{\mathbf{xx}}^{-1}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}\mathbf{a}_1 &= \lambda_1\mathbf{a}_1.\end{aligned} \tag{45}$$

In other words, $\mathbf{a}_1$ is the eigenvector with the maximum eigenvalue of the matrix $\Sigma_{\mathbf{xx}}^{-1}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}$. The subsequent pairs are found by using eigenvalues of decreasing magnitudes

$$\begin{aligned}\Sigma_{\mathbf{xx}}^{-1}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}} &= Q\Lambda Q^{-1} \\ &= \begin{bmatrix} | & | & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \\ | & | & | \end{bmatrix}\begin{bmatrix}\lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3\end{bmatrix}\begin{bmatrix} - & \mathbf{a}_1^T & - \\ - & \mathbf{a}_2^T & - \\ - & \mathbf{a}_3^T & -\end{bmatrix}\end{aligned} \tag{46}$$

where $|\lambda_1| \geq |\lambda_2| \geq |\lambda_3|$.

Because of the collinearity constraint, we have

$$\mathbf{b}_i \propto \Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}\mathbf{a}_i, i \in \{1, 2, 3\}. \tag{47}$$

The correlation coefficients are

$$\rho_i = \mathrm{Corr}(\mathbf{a}_i^T\mathbf{x}, \mathbf{b_i}^T\mathbf{y}), i \in \{1, 2, 3\}. \tag{48}$$

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For trace correlation derivation, according to (48)

$$\begin{aligned}
&\mathbf{a}_i^T\Sigma_{\mathbf{xx}}\mathbf{a}_i\mathbf{b}_i^T\Sigma_{\mathbf{yy}}\mathbf{b}_i\rho_i^2 = (\mathbf{a}_i^T\Sigma_{\mathbf{xy}}\mathbf{b}_i)^T\\
&\mathbf{a}_i^T\Sigma_{\mathbf{xx}}\mathbf{a}_i\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}\mathbf{a}_i^T\Sigma_{\mathbf{yy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}\mathbf{a}_i\rho_i^2\\
&\quad = (\mathbf{a}_i^T\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}\mathbf{a}_i)^T\\
&\Sigma_{\mathbf{xx}}^{-1}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}\mathbf{a}_i = \rho_i^2\mathbf{a}_i
\end{aligned} \tag{49}$$

which means ρ_i^2 is the corresponding eigenvalues. Thus

$$\lambda_i = \rho_i^2, i \in \{1,2,3\} \tag{50}$$

and

$$\begin{aligned}
\sum_{i=1}^{3}\rho_i^2 &= \mathrm{Tr}\left(\begin{bmatrix}\lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3\end{bmatrix}\right)\\
&= \mathrm{Tr}(Q^{-1}\Sigma_{\mathbf{xx}}^{-1}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}Q)\\
&= \mathrm{Tr}(\Sigma_{\mathbf{xx}}^{-1}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}Q^{-1}Q)\\
&= \mathrm{Tr}(\Sigma_{\mathbf{xx}}^{-1}\Sigma_{\mathbf{xy}}\Sigma_{\mathbf{yy}}^{-1}\Sigma_{\mathbf{yx}}).
\end{aligned} \tag{51}$$

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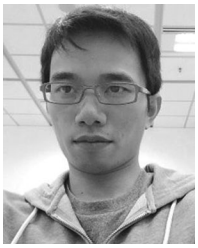
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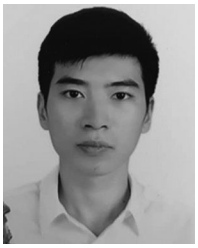
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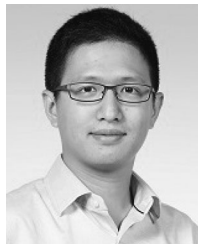
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